Practice Workbook §12.1 Simple Linear Regression

Question 1 (§12.1)

The model is $y = \beta_0 + \beta_1 x + \varepsilon$.

(a) Which part is the line of means? (b) What is ε? (c) State the four assumptions on ε.

Question 2 (§12.1)

Data (x, y): (1, 2), (2, 4), (3, 5), (4, 4), (5, 6).

Find $\bar{x}$, $\bar{y}$, Sxx and Sxy.

Question 3 (§12.1)

Same data. (a) Find b_1, b_0 and the least-squares line. (b) Predict y at x = 6.

Practice Workbook §12.2 An Analysis of Variance for Linear Regression

Question 1 (§12.2)

Data (x, y): (1, 2), (2, 4), (3, 5), (4, 4), (5, 6). Sxx = 10, Sxy = 8.

Find Total SS (Syy), SSR and SSE.

Question 2 (§12.2)

Same data. Complete the ANOVA table (source, df, SS, MS).

Question 3 (§12.2)

Same data. Estimate σ^2, the variance of ε, and σ.

Practice Workbook §12.3 Testing the Usefulness of the Regression Model

Question 1 (§12.3)

Same data: $b_1 = 0.8$, $s = 0.894$, $Sxx = 10$, $n = 5$.

Test $H_0: \beta_1 = 0$ against $H_a: \beta_1 \neq 0$ at $\alpha = .05$.

Question 2 (§12.3)

Same data. Find a 95% confidence interval for β_1.

Question 3 (§12.3)

Same data (SSR = 6.4, MSE = 0.8, Total SS = 8.8).

(a) Find the ANOVA F and compare with t^2. (b) Find r^2 and explain it.

Practice Workbook §12.6 Correlation Analysis

Question 1 (§12.6)

Data (x, y): (1, 2), (2, 4), (3, 5), (4, 4), (5, 6). Sxx = 10, Sxy = 8, Syy = 8.8.

Find the correlation coefficient r.

Question 2 (§12.6)

Same data. Test H_0: $\rho = 0$ against H_a: $\rho \neq 0$ at $\alpha = .05$.

Question 3 (§12.6)

Ice cream sales and drownings have r = 0.9 across months.

Does ice cream cause drowning? Explain.

Practice Workbook §13.1 The Multiple Regression Model

Question 1 (§13.1)

The model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$.

(a) How many β's? (b) What does β_1 mean?

Question 2 (§13.1)

$E(y) = 10 + 2x_1 + 5x_2$. (a) $E(y)$ at $x_1 = 3$, $x_2 = 1$? (b) At $x_1 = 3$, $x_2 = 0$? (c) What does the 5 mean?

Question 3 (§13.1)

A model has $k = 4$ predictors and $n = 30$ observations.

(a) How many parameters β? (b) Error df? (c) The four assumptions on ε?

Practice Workbook §13.2 Multiple Regression Analysis

Question 1 (§13.2)

n = 30, k = 2 predictors. SSR = 800, SSE = 270. Complete the ANOVA table.

Question 2 (§13.2)

Same output. Find R^2 and adjusted R^2.

Question 3 (§13.2)

Same output. Test H_0: $\beta_1 = \beta_2 = 0$ at $\alpha = .05$.

Practice Workbook §13.4 Quantitative and Qualitative Predictors

Question 1 (§13.4)

Cups sold y; x_1 = temperature (°C); x_2 = 1 if weekend, 0 if weekday.

Fitted: $\hat{y} = 13.42 + 3.50x_1 + 27.60x_2$.

Predict cups at 30 °C (a) on a weekday, (b) on a weekend.

Question 2 (§13.4)

Same model, n = 120. $b_2 = 27.60$ with SE = 2.37. Test H_0: $\beta_2 = 0$ at $\alpha = .05$.

Question 3 (§13.4)

Season has three levels: spring, summer, winter.

(a) How many dummy variables? (b) Define them.